

4.1 Vector Spaces and Subspaces

Definition

A vector space is a nonempty set V of objects, called vectors, on which are defined two operations, multiplication by scalars (real numbers) and addition, subject to the ten axioms listed below. The axioms must hold for all vectors $\vec{u}, \vec{v}$ and $\vec{w}$ in V and for all scalars c, d .

1. The sum of $\vec{u}$ and $\vec{v}$, $\vec{u} + \vec{v}$, is in V .
2. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$.
3. $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$.
4. There is a zero vector in V such that $\vec{u} + \vec{0} = \vec{u}$.
5. For each $\vec{u}$ in V , there is a vector $-\vec{u}$ in V such that $\vec{u} + (-\vec{u}) = \vec{0}$.
6. The scalar multiple of $\vec{u}$ by c , $c\vec{u}$, is in V .
7. $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$.
8. $(c+d)\vec{u} = c\vec{u} + d\vec{u}$.
9. $c(d\vec{u}) = (cd)\vec{u}$.
10. $1\vec{u} = \vec{u}$.

p. 226-227

I

subspaces

Definition

A subspace of a vector space V is a subset H of V that has three properties:

- a. The zero vector is in H .
- b. H is closed under vector addition. That is, for each $\vec{u}$ and $\vec{v}$ in H , the sum $\vec{u} + \vec{v}$ is in H .
- c. H is closed under scalar multiplication. That is, for each $\vec{u}$ in H and each scalar c , the vector $c\vec{u}$ is in H .

p. 229

A plane in $\mathbb{R}^3$ not through the origin is not a subspace of $\mathbb{R}^3$, because it does not contain the zero vector.

Similarly, a line in $\mathbb{R}^2$ not going through the origin is not a subspace of $\mathbb{R}^2$.

4.1 Vector Spaces and Subspaces (cont'd)

II

A subspace spanned by a set

Example 11

Given $\vec{v}_1$ and $\vec{v}_2$ in vector space V , let $H = \text{Span}\{\vec{v}_1, \vec{v}_2\}$. Show that H is a subspace of V .

Solution

The zero vector is in H , since $\vec{0} = 0\vec{v}_1 + 0\vec{v}_2$. To show that H is closed under vector addition, take two arbitrary vectors in H , say,

$$\vec{u} = s_1\vec{v}_1 + s_2\vec{v}_2 \quad \text{and} \quad \vec{w} = t_1\vec{v}_1 + t_2\vec{v}_2$$

By axioms 2, 3 and 8 for the vector space V ,

$$\begin{aligned}\vec{u} + \vec{w} &= (s_1\vec{v}_1 + s_2\vec{v}_2) + (t_1\vec{v}_1 + t_2\vec{v}_2) \\ &= (s_1 + t_1)\vec{v}_1 + (s_2 + t_2)\vec{v}_2\end{aligned}$$

So $\vec{u} + \vec{w}$ is in H . Furthermore, if c is any scalar, then by axioms 7 and 9

$$\begin{aligned}c\vec{u} &= c(s_1\vec{v}_1 + s_2\vec{v}_2) \\ &= (cs_1)\vec{v}_1 + (cs_2)\vec{v}_2\end{aligned}$$

which shows that $c\vec{u}$ is in H and H is closed under multiplication. H is a subspace of V .

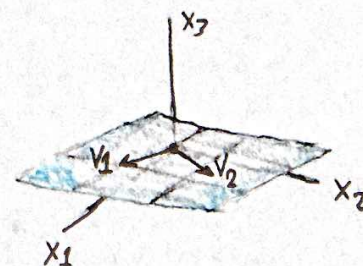


Figure: Example of a subspace.

Theorem 1

If $\vec{v}_1, \dots, \vec{v}_p$ are in a vector space V , then $\text{Span}\{\vec{v}_1, \dots, \vec{v}_p\}$ is a subspace of V .

Example 12

Let H be the set of all vectors of the form $(a-3b, b-a, a, b)$, where a and b are arbitrary scalars. That is, let

$$H = \{(a-3b, b-a, a, b) : a \text{ and } b \text{ in } \mathbb{R}\}.$$

Show that H is a subspace of $\mathbb{R}^4$.

Solution

Write the vectors in H as column vectors. Then an arbitrary vector in H has the form

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4.1 Vector Spaces and Subspaces (cont'd)

III

$$\begin{bmatrix} a-3b \\ b-a \\ a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$\uparrow$ $\uparrow$
 $\vec{v}_1$ $\vec{v}_2$

This calculation shows that $H = \text{Span}\{\vec{v}_1, \vec{v}_2\}$, where $\vec{v}_1$ and $\vec{v}_2$ are the vectors indicated above. Thus H is a subspace of $\mathbb{R}^4$ by theorem 1.

Example 13

For what value(s) of h will y be in the subspace of $\mathbb{R}^3$ spanned by $\vec{v}_1, \vec{v}_2, \vec{v}_3$ if

$$\vec{v}_1 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} \text{ and } y = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$$

Solution

Let $H = \text{Span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$, for y to be in H , it must be a linear combination of $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$.

Set up an augmented matrix and row reduce

$$\left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ -1 & -4 & 1 & 3 \\ -2 & -7 & 0 & h \end{array} \right]$$

$$\begin{aligned} R_2 &\leftarrow R_2 + R_1 \\ R_3 &\leftarrow R_3 + 2R_1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 3 & -6 & h-8 \end{array} \right]$$

$$R_3 \leftarrow R_3 - 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & (h-8)+3 \end{array} \right]$$

Cleaning up

$$\left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & h-5 \end{array} \right]$$

We see that $\vec{y}$ is in H for exactly $h=5$.